



WALMART

BUSINESS

SALES

ANALYSIS

Tools Used:MySQL | MySQL Workbench

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ABOUT WALMART

- **Foundation & Founder:** Walmart Inc. was founded by **Sam Walton** on **July 2, 1962**, in Arkansas, USA, and its current corporate headquarters is located in Bentonville, Arkansas.
- **Core Business:** The company is a retail giant that operates a vast network of hypermarkets, department stores, and e-commerce platforms to sell groceries, apparel, electronics, and household goods.
- **Method of Working:** It operates on an **"Everyday Low Prices" (EDLP)** strategy, leveraging bulk purchasing directly from manufacturers and highly efficient supply chain logistics to keep operational costs low.
- **Core Objective:** The official corporate mission statement of Walmart is **"Save Money. Live Better."**, which focuses on lowering the cost of living for consumers worldwide.
- **Financial Revenue:** It is the **world's largest company by revenue**, generating over **\$713 billion** annually according to recent fiscal disclosures.
- **Global Scale:** Walmart is the world's largest private employer with a workforce of **2.1 million associates**, managing more than **10,900 stores** across 19 countries.





PROJECT GOAL

The primary goal of this project was to transform a raw Walmart sales dataset into a clean and analysis-ready dataset using SQL.

The project was divided into two major phases :

Phase 1: Data Cleaning

- Identify and check null values
- Detect and remove duplicate records
- Validate and correct data types
- Ensure data consistency and quality

Phase 2: Business Analysis

After cleaning the dataset, SQL queries were used to answer key business questions related to:

- Revenue Performance
- Customer Behavior
- Profitability Analysis
- Time-Based Sales Trends

The final objective was to convert raw transactional data into meaningful business insights that can support data-driven decision making.

PROJECT WORKFLOW

Raw Dataset



Data Cleaning



Business Analysis



Insights Generation



Documentation



DATA CLEANING

Checked for Null Values

```
# CHECK THE NULL VALUES
SELECT * FROM walmart
WHERE invoice_id IS NULL
OR Branch IS NULL
OR City IS NULL
OR category IS NULL
OR unit_price IS NULL
OR quantity IS NULL
OR date IS NULL
OR time IS NULL
OR payment_method IS NULL
OR rating IS NULL
OR profit_margin IS NULL ;

# RESULT : NULL VALUES = NOT FOUND
```

Identified and Removed Duplicate Records

```
# CHECK THE DUPLICATE

SELECT
invoice_id ,COUNT(*) AS duplicate_count
FROM walmart
GROUP BY invoice_id
HAVING COUNT(*)>1;

CREATE TABLE temp_walmart AS
SELECT DISTINCT * FROM walmart
ORDER BY invoice_id;

TRUNCATE TABLE walmart;

INSERT INTO walmart
SELECT * FROM temp_walmart;

DROP TABLE temp_walmart;

SELECT invoice_id ,COUNT(*) FROM
walmart
GROUP BY invoice_id
HAVING COUNT(*) >1;

# REMOVE DUPLICATES SUCCESSFULLY !
```

Corrected Data Types

```
# CHANGE THE DATA TYPE :
SET SQL_SAFE_UPDATES = 0;

UPDATE walmart
SET unit_price = REPLACE(unit_price,'$', '');
ALTER TABLE walmart
MODIFY COLUMN unit_price FLOAT;

UPDATE walmart
) SET date = CASE
    WHEN date LIKE '%/_ _' THEN STR_TO_DATE(date, '%d/%m/%y')
    WHEN date LIKE '%/_ _ _ _' THEN STR_TO_DATE(date, '%d/%m/%y')
    ELSE date
- END;

UPDATE walmart
SET time = STR_TO_DATE(time, '%H:%i:%s');

ALTER TABLE walmart
MODIFY COLUMN date DATE;

ALTER TABLE walmart
MODIFY COLUMN time TIME;

SET SQL_SAFE_UPDATES = 1;

# DATA CLEANING IS COMPLETED SUCCESSFULLY!
```


Walmart BUSINESS QUESTIONS

Executive Overview

- Total Records
- Total Revenue
- Total Profit

Customer Behavior Analysis

- Highest Rated Category
- Overall Average Rating
- Average Quantity per Transaction
- Average Spend per Transaction
- Most Preferred Payment Method

Revenue Analysis

- Highest Revenue Category
- Highest Revenue City
- Top 3 Branches by Revenue
- Revenue Contribution (%) by Category

Profitability Analysis

- Most Profitable Category
- Most Profitable City

Time Analysis

- Highest Revenue Month
- Best Sales Time



SQL QUERIES

```
1 • USE walmart_db;
2
3 • SELECT * FROM walmart;
4
5 # ----- BUSINESS QUERIES -----
6
7 # ----- EXECUTIVE OVERVIEW -----
8
9 # TOTAL RECORDS :- 9969
10 • SELECT COUNT(*) FROM walmart;
11
12 # TOTAL REVENUE :- 1209726.38
13 • SELECT ROUND(SUM(unit_price*quantity),2) AS total_revenue FROM walmart;
14
15 # TOTAL PROFIT :- 476139.43
16 • SELECT ROUND(SUM(profit_margin*(unit_price*quantity)),2) AS total_profit FROM walmart;
17
18 # ----- REVENUE ANALYSIS-----
19
20 # HIGHEST REVENUE CATEGORY :- Fashion accessories
21 • SELECT category, ROUND(SUM(unit_price*quantity),2) AS REVENUE FROM walmart
22   GROUP BY category
23   ORDER BY REVENUE DESC
24   LIMIT 3;
25
26 # HIGHEST REVENUE CITY :- Weslaco
27 • SELECT City, ROUND(SUM(unit_price*quantity),2) AS REVENUE FROM walmart
28   GROUP BY City
29   ORDER BY REVENUE DESC
30   LIMIT 5;
```

```
31
32 # TOP 3 BRANCHES BY REVENUE :- WALM009, WALM074, WALM003
33 • SELECT Branch, ROUND(SUM(unit_price*quantity),2) AS REVENUE FROM walmart
34   GROUP BY Branch
35   ORDER BY REVENUE DESC
36   LIMIT 3;
37
38 # REVENUE CONTRIBUTION (%) BY CATEGORY :-Fashion accessories (40.46 %)
39 • SELECT category,
40       ROUND(SUM(unit_price * quantity), 2) AS revenue,
41       ROUND(
42           (SUM(unit_price * quantity) /
43            (SELECT SUM(unit_price * quantity) FROM walmart)) * 100,
44           2
45       ) AS revenue_percentage
46   FROM walmart
47   GROUP BY category
48   ORDER BY revenue_percentage DESC;
49
50 # ----- CUSTOMER BEHAVIOUR ANALYSIS -----
51
52 # HIGHEST RATED CATEGORY :- Food and beverages
53
54 • SELECT category ,ROUND(AVG(rating),1) AS AVG_RATINGS FROM walmart
55   GROUP BY category
56   ORDER BY AVG_RATINGS DESC;
57
58 # OVERALL AVERAGE RATINGS :- 5.8
59 • SELECT ROUND(AVG(rating),1) AS avg_rating FROM walmart ;
60
61 # AVERAGE QUANTITY :- 2.4
62 • SELECT ROUND(AVG(quantity),1) AS AVERAGE_QUANTITY FROM walmart;
63
64 # AVERAGE SPEND :- 121.35
65 • SELECT ROUND(AVG(unit_price*quantity),2) AS AVERAGE_SPEND FROM walmart;
66
67 # MOST PREFERRED PAYMENT METHOD :- CREDIT CARD
68 • SELECT payment_method,COUNT(payment_method) AS PAYMENT_COUNT FROM walmart
69   GROUP BY payment_method
70   ORDER BY PAYMENT_COUNT DESC;
71
```



SQL QUERIES

```
72 # ----- PROFIT ANALYSIS -----
73
74 # MOST PROFITABLE CATEGORY :- Fashion accessories (192314.89)
75
76 • SELECT category, ROUND(SUM((unit_price*quantity)*profit_margin),2) AS TOTAL_PROFIT FROM walmart
77   GROUP BY category
78   ORDER BY TOTAL_PROFIT DESC;
79
80 # MOST PROFITABLE CITY :- Weslaco(15296.09)
81 • SELECT City, ROUND(SUM((unit_price*quantity)*profit_margin),2) AS TOTAL_PROFIT FROM walmart
82   GROUP BY City
83   ORDER BY TOTAL_PROFIT DESC
84   LIMIT 5 ;
85
```

```
86 #----- TIME ANALYSIS-----
87
88 # HIGHEST REVENUE MONTH :- DECEMBER
89
90 • SELECT MONTHNAME(date) AS MONTHS,
91   ROUND(SUM(unit_price*quantity)) AS TOTAL_REVENUE FROM walmart
92   GROUP BY MONTHS
93   ORDER BY TOTAL_REVENUE DESC;
94
95 # BEST SELLING TIME :- EVENING (3PM-8:59PM)
96
97 • SELECT
98   CASE
99     WHEN HOUR(time) >=3 AND HOUR(time) <12 THEN 'MORNING'
100    WHEN HOUR(time) >=12 AND HOUR(time) <15 THEN 'AFTERNOON'
101    WHEN HOUR(time) >=15 AND HOUR(time) <21 THEN 'EVENING'
102   ELSE
103     'NIGHT'
104   END AS TIME_SLOT,
105   ROUND(SUM(unit_price*quantity),2) AS revenue
106   FROM walmart
107   GROUP BY TIME_SLOT
108   ORDER BY revenue DESC ;
109
110 # Business Analysis Completed Successfully !
```




KEY INSIGHTS

Revenue Insights

- Highest Revenue Category: **Fashion accessories**
- Highest Revenue City: **Weslaco**
- Top 3 Revenue Branches: **WALM009, WALM074, WALM003**

Customer Insights

- Highest Rated Category: **Food and beverages**
- Average Customer Rating: **5.8**
- Average Quantity Purchased: **2.4**
- Average Customer Spend: **121.35**
- Preferred Payment Method: **Credit Card**

Profit Insights

- Most Profitable Category: **Fashion accessories**
- Most Profitable City: **Weslaco**

Time Insights

- Highest Revenue Month: **December**
- Best Sales Time: **Evening (3PM-8:59PM)**



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